

\* Algebraists see structures: they study form and not content

## Discrete Groups & their reps:

Group:  $(G, \cdot)$   $\cdot$ : associative,  $\exists e$

$|G|$ : order / cardinality

\* Two groups are isomorphic  $G \cong H$  if the two groups are indistinguishable:

$\exists$  a group homo  $\varphi: G \rightarrow H$  that is bijective  
i.e.  $\varphi^{-1}: H \rightarrow G$  exists and is itself a group  
homo

Def: Automorphism: iso  $\varphi: G \rightarrow G$

ex:  $\mathbb{Z}_n \cong \{[0], [1], \dots, [n-1]\}$

integers mod  $n$  with  $\cdot \equiv +$  (addition)

ex:  $U_1(\mathbb{C}) \equiv \{z \in \mathbb{C} \mid |z| = 1\}$

ex:  $(\mathbb{R}^+, \cdot) \cong (\mathbb{R}, +)$

$\varphi \equiv \log$

• Note  $\text{Aut}(\mathbb{Z}) \cong \mathbb{Z}_2$

Corresponds to multiplying  $\mathbb{Z}$  by  $\pm 1$

$\Rightarrow \varphi: \mathbb{Z}_n \rightarrow U_1(\mathbb{C})$  is a group homomorphism  
 $\varphi([a]) := \exp(2\pi i a/n)$

•  $\text{Aut}(\mathbb{Z}_n) \cong \mathbb{Z}_n^\times$ ,  $(\mathbb{Z}_n^\times, \cdot)$

$\mathbb{Z}_n^\times \equiv$  all #'s  $1 \leq k \leq n$  coprime to  $n$

i.e.  $\gcd(k, n) = 1$

Fields: algebraic abstraction of the concept of a number

$\mathbb{C}, \mathbb{R}, \mathbb{Q}$

\*  $\mathbb{Z}_n$  is a field iff  $n$  is prime

$\mathbb{Z}_4$ : it is not possible to divide by  $[2]$  even though  $[2] \neq [0]$  there)

### Character of a field: $\mathbb{F}$

• Take some element  $K \in \mathbb{F}$ , then the character of  $\mathbb{F}$  is  $K$  iff

$$Kf = 0 \quad \forall f \in \mathbb{F}$$

ex:  $\mathbb{C}$  and  $\mathbb{R}$  have character 0 since

$$0 \cdot f = 0 \quad \forall f \in \mathbb{C} \text{ or } \mathbb{R}$$

•  $\mathbb{Z}_p = \{[0], \dots, [p-1]\}$  has character  $p$  since  
'  $[j] \in \mathbb{Z}_p$

$$\Rightarrow p[j] = [pj] = [jp] = j[p] = j[0] \\ = [j0] = [0]$$

as claimed.

\* Let  $H \subset G$ , then the index of  $H$  is the # of 'cosets' of  $H$

$\leadsto$  For finite groups:  $\text{Index}(H) = \frac{|G|}{|H|}$

A normal subgroup  $N \subset G$  is one which obeys

$$gNg^{-1} = N \quad \forall g \in G$$

$\leadsto$  The set  $G/H := \{[g]\}$

$$g \sim g' \Rightarrow gg^{-1} = h$$

i.e. the two elements are related by an element of  $h$ ;

$$g' = g^{-1}h_i = gh_i$$

$$\therefore \exists h_i \text{ s.t. } g' \sim g$$

Thus, for all  $H^0 \neq 1, \dots, |H|$

$g$  is related to elements  $g_i$

$$\{g_1 H, g_2 H, \dots\} \subset G/H$$

\*  $G/H$  has a group structure if  $H$  is normal

If  $H$  is a Normal subgroup of  $G$  then we write

$$H \triangleleft G \text{ (Normal Subgroup)}$$

$$H < G \text{ (Subgroup)}$$

Given a group homomorphism, the Kernel is defined as

$$\varphi: G \rightarrow H$$

$$\ker(\varphi) = \varphi^{-1}(e_H) = \{g \in G \mid \varphi(g) = e_H\}$$

Claim the  $\ker(\varphi)$  of  $\varphi: G \rightarrow H$  is a normal subgroup of  $G$ .

Proof:

$$\ker(\varphi) = \{k \in G \mid \varphi(k) = e\}$$

Take any  $g \in G$

(i)  $g = e$ : trivial

(ii)  $g \neq e$

$$\begin{aligned} \varphi(gkg^{-1}) &= \varphi(g)\varphi(k)\varphi(g^{-1}) \\ &= \varphi(g)e\varphi(g^{-1}) \\ &= \varphi(gg^{-1}) \\ &= \varphi(e) = e \end{aligned}$$

$$\therefore gkg^{-1} \in \ker(\varphi)$$

$$\exists m \in \ker(\varphi) \text{ s.t. } m = gkg^{-1}$$

Since  $k, g$  were arbitrary this holds for all  $k$ .

$$\therefore \ker(\varphi) \triangleleft G \quad \square$$

$$\text{Im}(\varphi) \cong G/\ker(\varphi) \mid \varphi: G \rightarrow H$$

A free group  $F_n$  with generators  $\{x_1, \dots, x_n\}$   
is the set of all possible "words" one can make  
w/  $x_1, x_1^{-1}, \dots, x_n, x_n^{-1}$

$e$ : empty word

ex:  $F_1 \cong \mathbb{Z}, \{1\}$

$\mathbb{Z}$  is generated by 1 via  $+$

$G$  is finitely generated if there are finitely many  
elements  $g_1, \dots, g_n \in G$  s.t.

$$G = \langle g_1, \dots, g_n \rangle$$

i.e. all  $g \in G$  can be written as some  
product(s) of  $g_i, g_j, g_k^{-1}$  etc....

ex: any finite group is finitely generated while  
 $\mathbb{R}$  is not

(cc) Any finitely gen. group  $G$  is the homomorphic image  
 $\varphi(F_n)$  of some free group  $F_n$   
 $G \cong F_n / \ker(\varphi)$

$$\varphi: F_n \rightarrow G$$

$$n=1: \varphi: \mathbb{Z} \rightarrow G$$

$\leadsto$  Leads to the idea of presentation

$$G \cong \langle X | R \rangle$$

$X$ : Set of generators of  $G$

$R$ : Set of relations

$R$ : words that equal the identity  $e$  in  $G$

• Enough words must be chosen so that  $\ker(\varphi)$  equals  
the smallest normal subgroup of  $F_n$  containing all  
of  $R$ .

ex: Dihedral group  $D_n$  (symmetries of the regular  
 $n$ -sided polygon)

$$D_n = \langle a, b \mid a^n = b^2 = abab = e \rangle$$

$a$ : rotations

b: reflections

Proof of  $abab$  for  $\Delta$ :  $b = \begin{matrix} \uparrow \\ \cdot \\ \downarrow \\ \cdot \\ \uparrow \end{matrix}$

$a: \begin{matrix} \cdot \\ \cdot \\ \cdot \end{matrix}$   $b: \begin{matrix} \cdot \\ \cdot \\ \cdot \end{matrix}$   $a: \begin{matrix} \cdot \\ \cdot \\ \cdot \end{matrix}$   $b: \begin{matrix} \cdot \\ \cdot \\ \cdot \end{matrix}$   $= e$

b holds for reflection along other vertices as well.

\* We say  $G$  equals (internal) direct product  $N \times H$

of subgroups if every element  $g \in G$  can be written UNIQUELY as a product  $nh \forall n \in N, h \in H$

$\Rightarrow$  If  $N$  and  $H$  are normal subgroups of  $G$

$$G = N \times H, \text{ if } \begin{matrix} N \trianglelefteq G \\ H \trianglelefteq G \end{matrix} \Leftrightarrow N \cap H = \{e\}$$

\* The (external) direct product  $N \times H$  is defined

to be an ordered pair  $(n, h)$  w/ operations

$$(n, h) \cdot (n', h') = (nn', hh')$$

$\Rightarrow G$  will be the internal product of its subgroups  $N, H$  iff it is isomorphic to their external direct product.

This implies

$$G \cong (N \times H)_{\text{ex}} \Rightarrow G \cong (N \times H)_{\text{int}}$$

Claim:  $N \cong G/H, H \cong G/N$

Proof: Since  $G$  is the internal product of  $N \times H$

$$g = nh \Leftrightarrow g \sim n$$
$$G/H = \{g_1 H, \dots, g_n H\}$$

Now,  $g_i \in N$  or  $g_i \in H$   
if  $g_i \in N$ , then

$$G/H = \{n_i H, \dots\}$$

for all  $g_i \in H$

$uH \in H$  trivially so only term  $e$  exists  
 $\{n_1H, \dots, n_{|N|}H, H\}$

Now, for

$$g_i = n_i H$$

but  $g_i$  is uniquely written as  $g_i = n_i \tilde{h}$   
 for some  $\tilde{h} \in H$

as such, this forces  $\tilde{h} = e$  for all remaining terms

$$G/H = \{n_1, \dots, n_{|N|-1}, e\} = N$$

as claimed

$$\therefore \boxed{G/H \cong N}$$

\* We say  $G$  is an (internal) semi-direct product  $N \rtimes H$  of subgroups if all conditions of the internal direct product are satisfied, except that  $H$  need not be normal in  $G$  but  $N \triangleleft G$  still.

The (external) semi-direct product  $N \rtimes_{\theta} H$  of two groups is defined to be all ordered pairs  $(n, h)$  w operation given by

$$\boxed{(n, h)(n', h') = (n \theta_h(n'), hh')}$$

where

$$\boxed{h \mapsto \theta_h \in \text{Aut}(N)}$$

can be any group homomorphism

Claim:  $N \rtimes_{\theta} H$  is a group under the operation above:

1c) "e":  $(n, h)e = (n, h)$

$$n = n \theta_h(n') \Rightarrow e = \theta_h(n')$$

But if  $\theta_h$  is an automorphism, then  $\exists \theta_h^{-1}$  s.t.

$$\theta_h^{-1}(e) = n'$$

but  $\forall \varphi \in \text{Aut } N \quad \varphi(e) = e$

$$\therefore \boxed{n' = e}$$

$$h = hh' \Rightarrow h' = e$$

$$\therefore \boxed{(e, e) = \tilde{e}} \text{ of } N \rtimes_{\theta} H$$

(ii) inverse elements:

$$(n, h)(n', h') = (n \ominus_h(n'), hh') = (e, e)$$

$$\boxed{h' = h^{-1}}$$

$$n \ominus_h(n') = e \Rightarrow \ominus_h(n') = n^{-1}$$

$$\boxed{\ominus_h^{-1}(n^{-1}) = n'}$$

$$\boxed{(n, h)^{-1} = (\ominus_h^{-1}(n^{-1}), h^{-1})}$$

(iii) Associativity:

$$\begin{aligned} & [(n, h)(n', h')](n'', h'') \\ &= (n \ominus_h(n'), hh')(n'', h'') \\ &= (n \ominus_h(n') \ominus_{hh'}(n''), hh'h'') \end{aligned}$$

$$\begin{aligned} (n, h)[(n', h')(n'', h'')] &= (n, h)(n' \ominus_{h'}(n''), h'h'') \\ &= (n \ominus_h(n' \ominus_{h'}(n'')), hh'h'') \\ &= (n \ominus_h(n') \ominus_h(\ominus_{h'}^{-1}(n'')), hh'h'') \end{aligned}$$

$\therefore$  In order for  $N \rtimes_{\ominus} H$  to be a group  $\ominus_h$  must have the following composition group structure

$$\boxed{\ominus_h \ominus_{h'} = \ominus_{hh'}}$$

This ensures associativity ✓

If  $\ominus_h = (\text{id})$  we recover the (external) direct product.

ex: Dihedral Group  $D_n \cong \mathbb{Z} \rtimes \mathbb{Z}_2$

(ii) Group of isometries in 3-space is

$$\mathbb{R}^3 \rtimes (\mathbb{Z}_2 \times \text{SO}(3))$$

$\mathbb{R}^3$ : translations

$\mathbb{Z}_2$ : reflections  $x \mapsto -x$  through the origin

$\text{SO}(3)$ : Rotations

\* Generally, if  $N$  is a normal subgroup of  $G$  ( $N \triangleleft G$ ) we say that  $G$  is an extension of  $N$  by <sup>the</sup> quotient group  $G/N$  equivalently,

if each element  $g \in G$  can be identified w a pair  $(n, h)$ , for  $n \in N$  and  $h \in H$ , and where

\* Group is abelian if  $gh = hg \quad \forall g, h \in G$

-  $\Rightarrow$  A group is cyclic if it has one generator:  $\mathbb{Z}_n, \mathbb{Z}$ .

The center  $Z(G) \triangleleft G$

$$Z(G) \equiv \{g \in G \mid ga = ag \quad \forall a \in G\}$$